Bose QuietComfort Ultra Headphones
Bluetooth SPP Protocol
Reverse-engineered

Krzysztof Kwiatkowski
https://docentcompany.com

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Version 2

Contents

1 Preamble

1.1 License

1.2 Motivation

Bose QuietComfort Ultra Headphones can only be configured using a dedicated Bose App available exclusively for mobile devices. This app requires iOS 17.0 or later or Android 13 and up (varies with device)[1]. If you do not have a smartphone that meets the requirements, you will not be able to use the headphones to their full potential. There is also no certainty that the application will be available in the future. In order to secure the future of headphone users and increase headphones potential, the communication protocol has been reverse engineered.

1.3 Method

The Bose App has been installed on the Android phone. With Bluetooth HCI snoop log enabled, actions in the app were perfomed. Logs from the device located in *bugreport.zip/FS/data/log/bt/btsnoop_hci.log* collected by `adb bugreport` command was opened in the Wireshark software. The logs were analyzed manually. Python was used to test the queries:

```python
import serial

ser = serial.Serial("COM4", 9600, timeout=1)
ser.write(bytes.fromhex("011b020101"))

data = ser.read(5) # Read 5 bytes
print(data.hex(" "), "".join(chr(b) if 32 <= b < 127 else "." for b in data))
```

Outgoing serial port can be found in the *More Bluetooth Settings -> COM Ports*.

Headphones firmware version: `1.6.7+g6ebabd2`.

[1] https://support.bose.com/s/article/quietcomfort-ultra-headphones-software-and-firmware-versions

2 Protocol description

Request and response header

hex:	XX XX	01 or 02 or 03 or 05 or 06 or 07	XX	...
desc:	Command	'Intentions': • **01** - Request sent by client to **get** informations. • **02** - Request sent by client to **set** some value in headphones. • **03** - Request sent by headphones as answer to the client request. • **04** - Request sent by headphones as answer to the client request. It means 'Command not found'. • **05** - Request sent by client when multiple requests as answer are expected (but single 06 can be returned). Cannot be replaced with 01 or 02. • **07** - Request sent by headphones as answer to the client request, that will be followed by another requests. • **06** - Request sent by headphones as answer to the client request, which means that this is the end of multiple answering requests.	Payload size	Payload

If command with intention `02` exists there is a there is a good chance that there is also a command with intent `01` with shared response.

2.1 Volume

2.1.1 Get current volume

Request

	4	3	2	1
hex:	05	05	01	00

Response

	6	5	4	3	2	1
hex:	05	05	03	02	1f	00 to 1f
desc:						Current volume

2.1.2 Set volume

Request

	5	4	3	2	1
hex:	05	05	02	01	00 to 1f
desc:					Volume to set

Response

The same as in the get current volume response.

2.2 Modes

Each command starts with **1f**.

2.2.1 Get current mode index

Request

	4	3	2	1
hex:	1f	03	01	00

Response

	5	4	3	2	1
hex:	1f	03	03	01	00 - 09
desc:					Mode index

2.2.2 Set current mode

Request

	6	5	4	3	2	1
hex:	1f	03	05	02	00 - 09	01
desc:					Mode index	???

Response

The same as in the get current mode response, but with the `06` intent.

2.2.3 Get informations about mode

Request

	5	4	3	2	1
hex:	1f	06	01	01	00 to 09
desc:					Mode index

Response

	51	50	49	48	47	46	45
hex:	1f	06	03	2f	00 - 09	00	01 - 22 or XX
desc:					mode index		Mode name spoken by headphones when changing the current mode via the button. 01 - Quiet, 02 - Aware, 03 - Transparent, 04 - Transparency, 05 - Masking, 06 - Comfort, 07 - Commute, 08 - Outdoor, 09 - Workout, 0a - Home, 0b - Work, 0c - Music, 0d - Focus, 0e - Relax, 0f - Flight, 10 - Airport, 11 - Driving, 12 - Training, 13 - Gym, 14 - Run, 15 - Walk, 16 - Hike, 17 - Talk, 18 - Call, 19 - Whisper, 1a - Hearing, 1b - Learn, 1c - Podcast, 1d - Audiobook, 1e - Conn (?), 1f - Sleep, 20 - Meditate, 21 - Yoga, 22 - Immersion, Others - No sound

	44	43	42
hex:	00 or 01	00	00 or 01
desc:	if the mode is created by the user	??	if the mode is favourite

	41 - 7
desc:	Ascii encoded mode name. For example: 51 75 69 65 74 (00)… for “Quiet”. “None” for not existing ones. Non-english characters are encoded incorrectly. TODO: Decode non-english chars.

	6	5	4	3	2	1
hex:	00 or 02 or 0d ??	00 - 0a	00 or 01	00 or 01 or 02	00 or 01	00 or 01
desc:	???	ANC level 00 is max level, 0a disabled	Is active sense enabled?	Immersive voice mode (0 - off, 1 - still, 2 - motion)	The same as byte no. 44	Is wind reduction enabled? If True, then ANC level is set to max (00)

Not existing / deleted mode looks like this:

	51	50	49	48	47	46	45	44	43	42	41 - 7	6	5	4	3	2	1
hex:	1f	06	03	2f	00 - 09	00	00	01	00	00	…	0d	0a	00	00	01	00
desc:					mode index						Ascii encoded: “None”						

2.2.4 Get list of modes

Request

	4	3	2	1
hex:	1f	01	05	00

Response

	4	3	2	1
hex:	1f	01	07	00

	9	8	7	6	5	4	3	2	1
hex:	1f	00	03	05	31	2e	30	2e	30
ascii:					1	.	0	.	0

	11	10	9	8	7	6	5	4	3	2	1
hex:	1f	02	03	07	03	07	00	00	00	1f	02

	5	4	3	2	1
hex:	1f	03	03	01	00 - 09 or ff
desc:					Curent mode index. If not existing mode (For e.g. quiet with immersive audio: ff).

	5	4	3	2	1
hex:	1f	05	03	01	00 or 01
desc:					Is remembering last mode when headphones are turned on on?

+ Repeated 10 times response in the same format as [the informations about mode](#).

	7	6	5	4	3	2	1
hex:	1f	08	03	03	0a	03	ff
desc:					??	??	??

	4	3	2	1
hex:	1f	01	06	00

2.2.5 Create mode

Updating mode is acomplished by creating a new one with the same mode index.

Request

	43	42	41	40	39	38	37
hex:	1f	06	02	27	03 - 09	00	01 - 22 or XX
desc:					mode index		Mode name spoken by headphones when changing the current mode via the button. 01 - Quiet, 02 - Aware, 03 - Transparent, 04 - Transparency, 05 - Masking, 06 - Comfort, 07 - Commute, 08 - Outdoor, 09 - Workout, 0a - Home, 0b - Work, 0c - Music, 0d - Focus, 0e - Relax, 0f - Flight, 10 - Airport, 11 - Driving, 12 - Training, 13 - Gym, 14 - Run, 15 - Walk, 16 - Hike, 17 - Talk, 18 - Call, 19 - Whisper, 1a - Hearing, 1b - Learn, 1c - Podcast, 1d - Audiobook, 1e - Conn (?), 1f - Sleep, 20 - Meditate, 21 - Yoga, 22 - Immersion, Others - No sound

	36 - 5
desc:	Ascii encoded mode name. For example: 52 65 6c 61 6b 73 (00)… for "Relaks". Non-english characters are encoded incorrectly. TODO: Decode non-english chars.

	4	3	2	1
hex:	00 - 0a	00	00 or 01 or 02	00 or 01
desc:	ANC level 00 is max level, 0a disabled	??	Immersive voice mode (0 - off, 1 - still, 2 - motion)	Enable wind reduction? If True, then ANC level should be set to max (00)

Response

The same as in the informations about mode.

2.2.6 Get favourites

Request

	4	3	2	1
hex:	1f	08	01	00

Response

	7	6	5	4	3	2	1
hex:	1f	08	03	03	0a	XX	XX
desc:						Bits are numerated from right. 1 means the mode with index corresponded to the bit number was set be favourite. 0 means set as not favourite.	

2.2.7 Set favourites

Request

	7	6	5	4	3
hex:	1f	08	02	03	0a

	2	1
bits:	0000 00XX	XXXX XXXX
desc:	Bits are numerated from right. 1 means the mode with index corresponded to the bit number will be favourite. 0 means not favourite.	

Response

The same as in the get favourites response.

2.2.8 Delete mode

Request

	5	4	3	2	1
hex:	1f	09	05	01	00 to 09
desc:					Index of the mode to delete

Response

	4	3	2	1
hex:	1f	09	06	00

2.2.9 Get remember last mode

Request

	4	3	2	1
hex:	1f	05	01	00

Response

	5	4	3	2	1
hex:	1f	05	03	01	00 or 01
desc:					If the last mode is remembered when the headphones are shut down.

2.2.10 Set remember last mode

Request

	5	4	3	2	1
hex:	1f	05	02	01	00 or 01
desc:					Should be last mode remembered when the headphones are shut down?

Response

The same as in the get remember last mode response.

2.3 Immersion mode

2.3.1 Get current immersion mode

Request

	4	3	2	1
hex:	05	0f	01	00

Response

	5	4	3	2	1
hex:	05	0f	03	01	00 or 01 or 02
desc:					Set immersive voice mode (0 - off, 1 - still, 2 - motion)

2.3.2 Set immersion mode

Request

	5	4	3	2	1
hex:	05	0f	02	01	00 or 01 or 02
desc:					Immersive voice mode to set (0 - off, 1 - still, 2 - motion)

Response

The same as in the get current immersion mode response.

2.3.3 Get current immersion mode calibration

Request

	4	3	2	1
hex:	05	11	01	00

Response

	102	101	100	99	98 - 1
hex:	05	11	03	62	…
desc:					Calibrated position in XYZ. TODO: Decode orientation representation.

2.3.4 Calibrate immersion mode

Request

The head should be pointed forward during calibration.

	4	3	2	1
hex:	05	11	05	00

Response

	4	3	2	1
hex:	05	11	07	00

	5	4	3	2	1
hex:	05	11	06	01	01 ??

2.4 Equalizer

2.4.1 Get equalizer settings

Request

	4	3	2	1
hex:	01	07	01	00

Response

	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
hex:	01	07	03	0c	f6	0a	00 - 0a or f6 - ff	00	f6	0a	00 - 0a or f6 - ff	01	f6	0a	00 - 0a or f6 - ff	02
desc:							Level for low tones; 00 - 0a for positive levels and f6 - ff for negative levels				Level for medium tones; 00 - 0a for positive levels and f6 - ff for negative levels				Level for high tones; 00 - 0a for positive levels and f6 - ff for negative levels	

2.4.2 Set equalizer settings

Request

	6	5	4	3	2	1
hex:	01	07	02	02	00 - 0a or f6 - ff	00 or 01 or 02
desc:					Value to set: 00 - 0a for positive ones and f6 - ff for negative ones	Value for which tones should be set: 0 - low, 1 - medium - 2 - high

Response
The same as for the get equalizer settings response.

2.5 Shortcut

2.5.1 Get current shortcut

Request

	4	3	2	1
hex:	01	09	01	00

Response

	11	10	9	8	7	6	5	4	3	2	1
hex:	01	09	03	07	80	09	0e or 03 or 13 or 01 or 10	00	09	40	0a
desc:							• 0e - disabled • 03 - battery level • 13 - immersion mode • 01 - voice assistant • 10 - spotify shortcut				

2.5.2 Set shortcut

Request

	7	6	5	4	3	2	1
hex:	01	09	02	03	80	09	0e or 03 or 13 or 01 or 10
desc:							• 0e - disabled • 03 - battery level • 13 - immersion mode • 01 - voice assistant • 10 - spotify shortcut

Response
The same as for the get current shortcut response.

2.6 Settings

2.6.1 Device name

The device name can be between 0 and 26 characters long.

2.6.1.1 Get device name

Request

	4	3	2	1
hex:	01	02	01	00

Response

hex:	01	02	03	00 - 1A	00	…
desc:						Ascii encoded device name

2.6.1.2 Set device name

Request

hex:	01	02	02	00 - 1A	…
desc:					Ascii encoded string to be set as a device name

Response

The same as the get device name response.

2.6.2 Level of microphone monitoring during calls

2.6.2.1 Get current level

Request

	4	3	2	1
hex:	01	0b	01	00

Response

	7	6	5	4	3	2	1
hex:	01	0b	03	03	01	00 or 01 or 02 or 03	0f
desc:						• 0 - off • 1 - low • 2 - medium • 3 - high	

2.6.2.2 Set level

Request

	6	5	4	3	2	1
hex:	01	0b	02	02	01	00 or 01 or 02 or 03
desc:						• 0 - off • 1 - low • 2 - medium • 3 - high

Response

The same as for the get level of microphone monitoring during calls response.

2.6.3 Auto off time

2.6.3.1 Get current auto off time

Request

	4	3	2	1
hex:	01	04	01	00

Response

	7	6	5	4	3	2	1
hex:	01	04	03	03	XX	00	XX
desc:					• 00 00 00 - never • 05 00 00 - 5 minutes • 14 00 00 - 20 minutes • 28 00 00 - 40 minutes • 3c 00 00 - 1 hour • b4 00 00 - 3 hours • a0 00 05 - 24 hours		

2.6.3.2 Set auto off time

Request

hex:	01	04	02	01 or 02	XX or XX XX
desc:					• 00 - never • 05 - 5 minutes • 14 - 20 minutes • 28 - 40 minutes • 3c - 1 hour • b4 - 3 hours • a0 05 - 24 hours

Response
The same as in response of get auto off time.

2.6.4 Auto stopping music when headphones are not on the head

2.6.4.1 Get current setting

Request

	4	3	2	1
hex:	01	18	01	00

Response

	5	4	3	2	1
hex:	01	18	03	01	00 or 01
desc:					Is music stopped when headphones are not on head? 1 - yes, 0 - no

2.6.4.2 Set auto stopping music when headphones are not on the head

Request

	5	4	3	2	1
hex:	01	18	02	01	00 or 01
desc:					Should music be stopped when headphones are not on head? 1 - yes, 0 - no

Response
The same as for the get auto stopping music when headphones are not on the head response.

2.6.5 Auto call answering when headphones are put on head

2.6.5.1 Get current setting

Request

	4	3	2	1
hex:	01	1b	01	00

Response

	5	4	3	2	1
hex:	01	1b	03	01	00 or 01
desc:					Are calls auto answered when headphones are put on head? 1 - yes, 0 - no

2.6.5.2 Set auto call answering when headphones are put on head

Request

	5	4	3	2	1
hex:	01	1b	02	01	00 or 01
desc:					Should calls be auto answered when headphones are put on head? 1 - yes, 0 - no

Response
The same as for the get auto call answering when headphones are put on head response.

2.6.6 Voice prompts

2.6.6.1 Get options

Request

	4	3	2	1
hex:	01	03	01	00

Response

	11	10	9	8
hex:	1f	03	03	07

	7			
bits:	0	0 or 1	0 or 1	0 0000
desc:		Is current language a default one (english)? 0 - no, 1 - yes	Are voice prompts about connected devices enabled? 0 - no, 1 - yes	Set language of voice prompts. • 0 0001 - English • 0 0010 - French • 0 0011 - Italian • 0 0100 - German • 0 0110 - Spanish • 0 1000 - Chinese • 0 1111 - Guangdong dialect; Cantonese • 1 0000 - Japanese

	6	5	4	3	2	1
hex:	00	01	81	e5	01	00 or 01
desc:						Is battery level info enabled when headphones are turn on? 0 - no, 1 - yes

2.6.6.2 Set options

Request

	6	5	4	3
hex:	01	03	02	02

	2			
bits:	0	0	0 or 1	0 0000
desc:			Should voice prompts about connected devices be enabled? 0 - no, 1 - yes	Set language of voice prompts. • 0 0001 - english • 0 0010 - french • 0 0011 - italian • 0 0100 - german • 0 0110 - spanish • 0 1000 - Chinese • 0 1111 - Guangdong dialect; Cantonese • 1 0000 - Japanese

	1
hex:	00 or 01
desc:	Should battery level info be enabled when headphones are turn on? 0 - no, 1 - yes

Response

The same as in response of get voice prompts options.

2.7 Bluetooth

2.7.1 Multipoint connections

2.7.1.1 Are enabled?

Request

	4	3	2	1
hex:	01	0a	01	00

Response

	5	4	3	2
hex:	01	0a	03	01

	1							
bits:	0	0	0	0	0	1	1	0 or 1
desc:								Are multipoint connections enabled? 0 - no, 1 - yes

2.7.1.2 Turn on and off multipoint connection

Request

	5	4	3	2	1
hex:	01	0a	02	01	00 or 01
desc:					Should multipoint connections be enabled? 0 - no, 1 - yes

Response
Teh same as for the [are multipoint connections enabled response](#).

2.7.2 Get paired devices identificator

Request

	4	3	2	1
hex:	04	04	01	00

Response

hex:	04	04	03	k*6 + 1	03	…	…	…
desc:				Every paired device identificator is 6 bytes long		…	Second (?) paired device identificator	First (?) paired device identificator

2.7.3 Get paired device informations

Request

	10	9	8	7	…
hex:	04	05	01	06	…
desc:					Paired device identificator

Response

hex:	04	05	03	XX	64	89	F1	26	8e	8e	00 or 01 or 03??	02	03	...
desc:				Payload size	?	?	?	?	?	?	Changes if device is connected or disconnected	?	?	Paired device name

2.8 Informations

2.8.1 Get current playing media informations

Request

	4	3	2	1
hex:	05	06	05	00

Response

	4	3	2	1
hex:	05	06	07	00

hex:	05	06	03	XX	00	...
desc:						Ascii encoded current playing media title

hex:	05	06	03	XX	01	...
desc:						Ascii encoded current playing media artist

	4	3	2	1
hex:	05	06	06	00

2.8.2 Get headphones firmware version

Request

	4	3	2	1
hex:	00	05	01	00

Response

hex:	00	05	03	XX	...
desc:					Ascii encoded current firmware version. For e.g. ‘1.6.7+g6ebabd2’.

2.8.3 Get headphones serial number

Request

	4	3	2	1
hex:	00	07	01	00

Response

hex:	00	07	03	XX	...
desc:					Ascii encoded serial number.

2.8.4 Get headphones product identification number (GUID)

Request

	4	3	2	1
hex:	00	0C	01	00

Response

	20	19	18	17	16 - 1
hex:	00	0C	03	10	...
desc:					GUID. NOT ascii encoded.

2.8.5 Get ???? version

Request

	4	3	2	1
hex:	00	01	01	00

Response

hex:	02	00	03	XX	...
desc:					Ascii encoded version. For e.g.: '1.2.0'

2.8.6 Get ???? version

Four possible requests with four possible responses. All returned values expect the first one and the last one is the same for all requests.

Request

	4	3	2	1
hex:	02	01	05	00

	4	3	2	1
hex:	01	01	05	00

	4	3	2	1
hex:	00	04	05	00

	4	3	2	1
hex:	05	02	05	00

Response

	4	3	2	1
hex:	02	01	07	00

	4	3	2	1
hex:	01	01	07	00

	4	3	2	1
hex:	00	04	07	00

	4	3	2	1
hex:	05	02	07	00

hex:	02	00	03	XX	...
desc:					Ascii encoded version. For e.g.: '1.1.0'

	8	7	6	5	4	3	2	1
hex:	02	02	03	04	28	FF	FF	00
desc: ????								

	5	4	3	2	1
hex:	02	05	03	01	00
desc: ????					

	8	7	6	5	4	3	2	1
hex:	02	0F	03	04	00	60	00	00
desc: ????								

	5	4	3	2	1
hex:	02	10	03	01	01
desc: ????					

	5	4	3	2	1
hex:	02	11	03	01	01
desc: ????					

	4	3	2	1
hex:	02	01	06	00

	4	3	2	1
hex:	01	01	06	00

	4	3	2	1
hex:	00	04	06	00

	4	3	2	1
hex:	05	02	06	00

Appendix

Advanced debugging Python code

```python
import serial
import time

PORT = "COM4"


def send_and_read(hex_string):
    time.sleep(0.15)
    tx_bytes = bytes.fromhex(hex_string)

    with serial.Serial(PORT, 9600, timeout=1) as ser:
        ser.write(tx_bytes)
        print()
        print(f"Sent: {tx_bytes.hex().upper()}")

        was_seven = False
        while True:
            # Read header
            header = ser.read(4)
            if len(header) < 4:
                print("No full response")
                return

            cmd1, cmd2, status, length = header
            payload = ser.read(length)

            if len(payload) < length:
                print("Invalid payload")
                return

            frame = header + payload

            # Display HEX
            print("\n--- Recived frame ---")
            print(f"HEX:   {frame.hex(" ").upper()}")

            # Display ASCII
            ascii_repr = "".join(chr(b) if 32 <= b <= 126 else "." for b in frame)
            print(f"ASCII: {ascii_repr}")

            # Recive
            if status == 0x06:
                print("STATUS 6 -> end of the response")
                break

            if status == 0x07:
                was_seven = True
                print("STATUS 7 -> await more responses")
                continue

            if status == 0x03:
                if was_seven:
                    print("STATUS 3 -> no more responses but wait for 6")
```

```
                continue
            else:
                break
                # continue

        print(f"Unkown status {status:02X}, stopped")
        break

    print("--- --- ---")
```

Changes

Version 2

- Fixed typo in the get informations about mode header.
- Fixed typo in the create mode section.
- Fixed an incorrect link to the response fomat for create mode.
- Added discovered mode names that spoken by the headphones when changing the current mode via button for both get informations about mode and create mode.
- Added byte indexes for the get current immersion mode calibration response.
- Added byte indexes for the get headphones product identification number response.
- Fixed too wide table in the get paired devices identificator request.
- Added the maximum device name length.
- Changed get paired devices identificator response to support a variable number of paired devices.
- Added mode info response schema for not existing / deleted modes.
- Changed header title from “Get current time settings” to “Get current auto off time”.
- Added get current mode index endpoint.
- Changed set current mode position in the docs (which means that header numbers changed).
- Fixed typo from “0ff” to “off” in the document.